Ming-Yang Zhuang

Curriculum Vitae

Department of Astronomy
University of Illinois Urbana-Champaign
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Education & Work

2022–present **Postdoctoral Research Scholar**, *University of Illinois Urbana-Champaign*, Urbana, IL, USA (advisor: Yue Shen).

2016–2022 **PhD, Astrophysics**, *Peking University*, Beijing, China (advisor: Luis C. Ho).

Mentoring Experience

2025-present Undergraduate student: Ferdinand Ferdinand in DoA UIUC

2024-present Graduate students: Liam Nolan (co-mentored with Xin Liu; mentored paper #4) in DoA UIUC

2022-2024 Graduate students: Wen Sun (mentored paper #2) and Changhao Chen (mentored paper #3) in KIAA co-mentored with Luis C. Ho

2020-2022 Undergraduate student: Zihao Wu (mentored paper #1) in DoA PKU co-mentored with Luis C. Ho (now in CfA/Harvard)

Teaching Experience & Service Work

2026 NASA ADAP25 review panelist

2025 JWST summer school lecturer (Westlake University)

2024 JWST Cycle 4 Executive Committee external reviewer

2023-present JCMT proposal reviewer

2023 JWST Cycle 3 discussion panelist (SMBH2)

2022-2023 CSST summer school lecturer (Peking University)

2020-present AAS journal Reviewer

Research & Interests

- Galaxy formation and evolution
- Coevolution of supermassive black holes and their host galaxies
- AGN outflows and feedbacks
- High-z tidal disruption events and transients

Approved Proposal

2025 JWST Cycle 4 (GO 7893; PI), Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

2024 JWST Cycle 3 (GO 5105; Co-PI), NEXUS: the North ecliptic pole EXtragalactic Unified Survey

2024 JWST Cycle 2 (DD 6595; PI), Giant in the Infant Universe: Identification of A Record-breaking Strongly-lensed (Ancestor of) Extremely Massive Quiescent Galaxy

2023 JWST Cycle 2 (GO 3038; PI), An archival study of AGN host galaxies with JWST NIRCam imaging

Selected Conferences

2024 "NEXUS: the North ecliptic pole EXtragalactic Unified Survey", *The Ecosystem of Gas and Dust in Galaxies Near and Far 2024*, Chiang Mai, Thailand. **(invited talk)**

2024 "Evolutionary Paths of Active Galactic Nuclei and Their Host Galaxies", *Workshop on Galaxy and Black-hole evolution*, Kavli IPMU, Tokyo, Japan. (contributed talk)

2023 "Characterization of JWST NIRCam PSFs and Implications for AGN+Host Image Decomposition", *Roman Science Inspired by JWST Results Conference*, STScI, Baltimore, USA. (contributed talk)

2021 "2D Imaging Analysis", *CSST Sky Survey Image Analysis Forum*, Chongqing, China. **(invited talk)**

2020 "The Interplay Between Star Formation and Black Hole Accretion in Nearby Active Galaxies", *Supermassive Black Hole Academic Forum*, Lijiang, China. **(invited talk)**

2018 "The Infrared Emission and Opening Angle of the Torus in Quasars", *2018 PUC-KIAA Workshop*, Santiago, Chile. (contributed talk)

Publications

First-Author Papers (14)

14. Zhuang, M.-Y., & Shen, Y., & Li, J., et al. "NEXUS: Quick Release Notes", 2026, arXiv:2603.04586
13. Zhuang, M.-Y., & Shangguan, J., & Bian, Y., et al. "Cold Gas Infall onto A Brightest Group Galaxy via A Gas-Rich Minor Merger", 2026, **ApJ**, 999, 16
12. Zhuang, M.-Y., & Li, J., & Shen, Y., et al. "NEXUS: A Spectroscopic Census of Broad-line AGNs and Little Red Dots at $3 < z < 6$", 2026, **ApJ**, 999, 31
11. Zhuang, M.-Y., & Wang, F., & Sun, F., et al. "NEXUS Early Data Release: NIRCam Imaging and WFSS Spectroscopy from the First (Partial) Wide Epoch", 2024, **ApJS**, 282, 54
10. Zhuang, M.-Y., & Yang, Q., & Shen, Y., et al. "High-quality Extragalactic Legacy-field Monitoring (HELM) with DECam", 2024, **ApJS**, 274, 42
9. Zhuang, M.-Y., & Shen, Y. "Characterization of JWST NIRCam PSFs and Implications for AGN+Host Image Decomposition", 2024, **ApJ**, 962, 139

8. Zhuang, M.-Y., Li, J., & Ho, L. C. "AGNs and Host Galaxies in COSMOS-Web. I. NIRCam Images, PSF Models and Initial Results on X-ray-selected Broad-line AGNs at $0.35 < z < 3.5$", 2024, **ApJ**, 962, 93
7. Zhuang, M.-Y., & Ho, L. C. "Evolutionary paths of active galactic nuclei and their host galaxies", 2023, **Nature Astronomy**, 7, 1376
6. Zhuang, M.-Y., & Ho, L. C. "The Star-forming Main Sequence of the Host Galaxies of Low-redshift Quasars", 2022, **ApJ**, 934, 130
5. Zhuang, M.-Y., Ho, L. C., & Shangguan, J. "Black Hole Accretion Correlates with Star Formation Rate and Star Formation Efficiency", 2021, **ApJ**, 906, 38
4. Zhuang, M.-Y., & Ho, L. C. "The Interplay Between Star Formation and Black Hole Accretion in Nearby Active Galaxies", 2020, **ApJ**, 896, 108
3. Zhuang, M.-Y., Ho, L. C., & Shangguan, J. "New Method to Measure Star Formation Rates in Active Galaxies Using Mid-infrared Neon Emission Lines", 2019, **ApJ**, 873, 103
2. Zhuang, M.-Y., & Ho, L. C. "Recalibration of [O II] $\lambda 3727$ as a Star Formation Rate Estimator for Active and Inactive Galaxies", 2019, **ApJ**, 882, 89
1. Zhuang, M.-Y., Ho, L. C., & Shangguan, J. "The Infrared Emission and Opening Angle of the Torus in Quasars", 2018, **ApJ**, 862, 118

(Co-)Mentored Papers (5)

5. Vizgan, D., **Zhuang, M.-Y.**, Smail, I., et al. "JWST's PEARLS: A clumpy ring galaxy at z=4.0148", 2026, **ApJ** accepted, arXiv:2603.11575
4. Nolan, L., **Zhuang, M.-Y.**, Liu, X., et al. "Frequency and Abundance of Binary sUpermassive bLack holes from Optical Variability Surveys (FABULOVS): Hubble Space Telescope Imaging of Radial-Velocity Selected Binary Candidates", 2025, **ApJ** accepted, arXiv:2505.10805
3. Chen, C.-H., Ho, L. C., Li, R., **Zhuang, M.-Y.** "The Host Galaxy (If Any) of the Little Red Dots", 2025, **ApJ**, 983, 60
2. Sun, W., Ho, L. C., **Zhuang, M.-Y.**, et al. "The Structure and Morphology of Galaxies during the Epoch of Reionization Revealed by JWST", 2024, **ApJ**, 960, 104
1. Wu, Z., Ho, L. C., & **Zhuang, M.-Y.** "An Elusive Population of Massive Disk Galaxies Hosting Double-lobed Radio-loud Active Galactic Nuclei", 2022, **ApJ**, 941, 95

Contributed Papers (30) [see this ADS link for the most up-to-date list]